

1. The code defines four operation labels: "somma" for addition, "sottrazione" for subtraction, "molt" for multiplication, and "div" for division. Each of these labels performs the corresponding operation using the values in s1 and s2 and stores the result in s3.

- **somma**: Adds s1 and s2 and stores the result in s3.
- **sottrazione**: Subtracts s2 from s1 and stores the result in s3.
- **molt**: Multiplies s1 and s2 and stores the result in s3.
- **div**: Divides s1 by s2 and stores the result in s3.

Addition:

8	s0/fp	1	00000000000000000000000000000000 00000000000000000000000000000001
9	s1	5	00000000000000000000000000000000 00000000000000000000000000000101
10	a0	0	00000000000000000000000000000000 00000000000000000000000000000000
11	a1	0	00000000000000000000000000000000 00000000000000000000000000000000
12	a2	0	00000000000000000000000000000000 00000000000000000000000000000000
13	a3	0	00000000000000000000000000000000 00000000000000000000000000000000
14	a4	0	00000000000000000000000000000000 00000000000000000000000000000000
15	a5	0	00000000000000000000000000000000 00000000000000000000000000000000
16	a6	0	00000000000000000000000000000000 00000000000000000000000000000000
17	a7	0	00000000000000000000000000000000 00000000000000000000000000000000
18	s2	10	00000000000000000000000000000000 00000000000000000000000000001010
19	s3	15	00000000000000000000000000000000 00000000000000000000000000001111

Subtraction:

8	s0/fp	2	00000000000000000000000000000000 00000000000000000000000000000010
9	s1	5	00000000000000000000000000000000 00000000000000000000000000000101
10	a0	0	00000000000000000000000000000000 00000000000000000000000000000000
11	a1	0	00000000000000000000000000000000 00000000000000000000000000000000
12	a2	0	00000000000000000000000000000000 00000000000000000000000000000000
13	a3	0	00000000000000000000000000000000 00000000000000000000000000000000
14	a4	0	00000000000000000000000000000000 00000000000000000000000000000000
15	a5	0	00000000000000000000000000000000 00000000000000000000000000000000
16	a6	0	00000000000000000000000000000000 00000000000000000000000000000000
17	a7	0	00000000000000000000000000000000 00000000000000000000000000000000
18	s2	10	00000000000000000000000000000000 00000000000000000000000000001010
19	s3	-5	11111111111111111111111111111111 11111111111111111111111111111011

Multiplication:

8	s0/fp	3	00000000000000000000000000000000 00000000000000000000000000000011
9	s1	5	00000000000000000000000000000000 00000000000000000000000000000101
10	a0	0	00000000000000000000000000000000 00000000000000000000000000000000
11	a1	0	00000000000000000000000000000000 00000000000000000000000000000000
12	a2	0	00000000000000000000000000000000 00000000000000000000000000000000
13	a3	0	00000000000000000000000000000000 00000000000000000000000000000000
14	a4	0	00000000000000000000000000000000 00000000000000000000000000000000
15	a5	0	00000000000000000000000000000000 00000000000000000000000000000000
16	a6	0	00000000000000000000000000000000 00000000000000000000000000000000
17	a7	0	00000000000000000000000000000000 00000000000000000000000000000000
18	s2	10	00000000000000000000000000000000 000000000000000000000000000001010
19	s3	50	00000000000000000000000000000000 0000000000000000000000000000110010

Division:

8	s0/fp	4	00000000000000000000000000000000 00000000000000000000000000000100
9	s1	50	00000000000000000000000000000000 0000000000000000000000000000110010
10	a0	0	00000000000000000000000000000000 00000000000000000000000000000000
11	a1	0	00000000000000000000000000000000 00000000000000000000000000000000
12	a2	0	00000000000000000000000000000000 00000000000000000000000000000000
13	a3	0	00000000000000000000000000000000 00000000000000000000000000000000
14	a4	0	00000000000000000000000000000000 00000000000000000000000000000000
15	a5	0	00000000000000000000000000000000 00000000000000000000000000000000
16	a6	0	00000000000000000000000000000000 00000000000000000000000000000000
17	a7	0	00000000000000000000000000000000 00000000000000000000000000000000
18	s2	10	00000000000000000000000000000000 000000000000000000000000000001010
19	s3	5	00000000000000000000000000000000 00000000000000000000000000000101

2.

a. add x9, x20, x21

0000000	10101	10100	000	01001	0110011
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b. addi x10, x22, 36

000000100100	10110	000	01010	0010011
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3. **Single Precision:**

Exponent (8 bits) = $6 + 127 = 133 = 10000101$

Fraction (23 bits) = 0000 0111 0000 0000 000

Final bit pattern (32 bits): 1 1000 0101 0000 0111 0000 0000 0000 000

Double Precision:

Exponent (11 bits) = $6 + 1023 = 1029 = 100000000101$

Fraction (52 bits) = 0000 0111 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000

Final bit pattern (64 bits): 1 1000 0000 1010 0000 1110 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000

4.

$0x40000000 = 0\ 10000000\ 000000000000000000000000 = 2.0$

$0x427D0000 = 010000100\ 111110100000000000000000 = 63.25$

add: $0x42828000 = 0\ 10000101\ 000001010000000000000000 = 65.25$

mul: $0x42fd0000 = 0\ 10000101\ 111110100000000000000000 = 126.5$

The results are correct.

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1  addi sp, sp, -8
2  lui x5, 0x427D0
3  lui x6, 0x40000
4  sw x5, 0(sp)
5  sw x6, 4(sp)
6  flw f0, 4(sp)
7  flw f1, 0(sp)
8
9  fadd.s f2, f0, f1
10 fmul.s f3, f0, f1
```

Name	Number	Value
ft0	0	0x40000000
ft1	1	0x427d0000
ft2	2	0x42828000
ft3	3	0x42fd0000